

Technical Report on Image Segmentation using Mask2Former

Mayukh Kundu

I. ABSTRACT

This paper explores the use of the Mask2Former model for semantic segmentation. Specifically, addressing the challenges in Indian road datasets under various scenarios. Mask2Former’s mask focus mechanism and multi-scale high-resolution feature enhance segmentation performance, especially in detecting rare classes. Evaporation studies show the effects of various parameters on segmentation quality. It emphasizes the adaptability of the model to different tasks. We evaluate our approach using metrics such as the average Dice coefficient, F1 score, and Mean IoU.

Index Terms—Image Segmentation, Mask2Former, Semantic Segmentation, Deep Learning, Transformer, Ablation Study.

II. INTRODUCTION

In computer vision, image segmentation is a basic operation. It assigns each pixel to a class and divides it into meaningful segments. They are used in fields ranging from autonomous driving to medical imaging, where accurate, real-time segmentation is important. Traditional methods for image segmentation usually rely on convolutional neural networks (CNNs) for per-pixel classification, but CNNs face challenges in capturing global context due to the limited receptive field. Recent advances use transformer-based architectures to exploit attention mechanisms. Mask2Former is a universal segmentation model designed to handle multiple semantic, instance, and aggregate segmentation tasks within a single architecture. Key innovations include a masked attention mechanism that focuses only on relevant areas, improving computational efficiency and accuracy. This paper uses Mask2Former on a large Indian road dataset with diverse environments and rare classes. The aim is to demonstrate the flexibility and effectiveness of the model in complex real-world situations.

III. APPROACH OVERVIEW

This approach focuses on optimizing Mask2Former for high-resolution image segmentation tasks, specifically addressing unbalanced rare classes in the dataset and also performing an ablation study for hyperparameter tuning.

A. Dataset Preprocessing

The dataset consists of high-resolution images (1920x1080) and corresponding segmented maps. The classes are very unbalanced; some represent thousands of samples, while others are rare. Pre-processing includes segmentation map analysis, identification of rare classes, and filtering out mislabeled samples. A custom extraction strategy was used by choosing images according to class variety, the existence of rare classes, and per-subset size to ensure balanced subsets for model fine-tuning.

B. Model Architecture

Mask2Former’s architecture is based on the Transformer framework. It is optimized for segmentation through multi-scale mask study and high-resolution feature extraction. Masked attention focuses on pixels in the projected mask area, ignoring unrelated regions and reducing computational burden. This targeted approach improves performance and enables fine-grained segmentation in complex scenes. The model makes use of a core feature extractor, a pixel decoder for high-resolution embedding, and a Transformer decoder that dynamically adjusts the segmentation.

C. Training and Fine-tuning

The training uses a pre-trained Mask2Former model, where the encoder and pixel decoder are frozen to preserve the pre-learned representation. Fine-tuning focuses on the Transformer decoder and the classification layer optimized for the current dataset. A custom data pipeline processes the input images into a Mask2Former-compatible format, generating segmentation maps and mask labels for each set. Training parameters include a learning rate of 5e-5, a batch size of 8, and an optimizer like Adam or AdamW for three epochs.

II. EQUATIONS

$$\text{Mean IoU} = \frac{1}{n} \sum_{i=1}^n \frac{|A_i \cap B_i|}{|A_i \cup B_i|} \quad (1)$$

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2)$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (3)$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (4)$$

$$\text{Dice Coefficient} = \frac{2 \times |A \cap B|}{|A| + |B|} \quad (5)$$

V. RESULTS AND DISCUSSION

The model's performance was evaluated using metrics namely mean Dice coefficient, F1 score, and the average IoU across validation images. The segmentation maps generated by Mask2Former demonstrated consistent accuracy, with high precision in rare class segmentation due to the model's ability to handle high-resolution features.

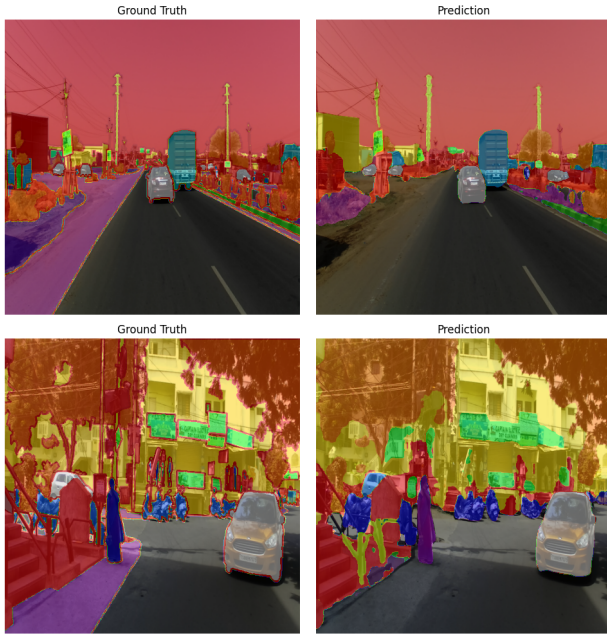


Fig. 1. Sample segmentation results on Indian roadway dataset showing segmentation for diverse classes with a comparison between Ground Truth and Prediction.

VI. ABLATION STUDY

Ablation studies are performed to understand the effects of key model components and parameters. Concealed attention and multi-scale features are the main contributions to the model's performance. It has a high-resolution feature map, which provides better segmentation accuracy for small objects. The following evaporation figures show the impact of different architectures. Performing Further Ablation studies by varying parameters such as Batch Size and Epochs and changing the Optimizer for Different Parameters, we obtained the Following Results.

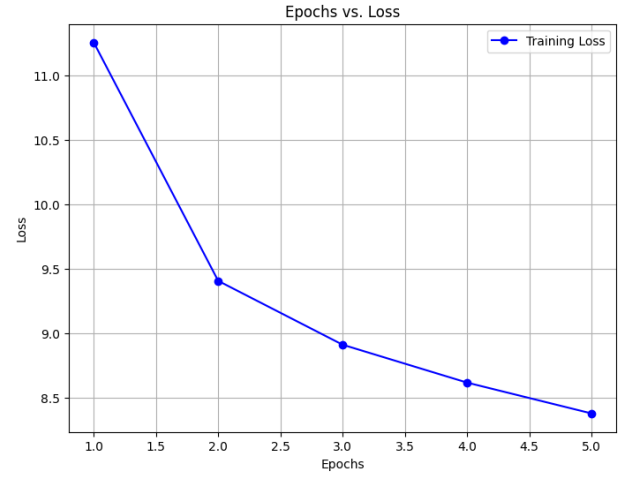


Fig. 2. Comparison of Training Loss Across Epochs for modified Mask2Former Image Segmentation Model

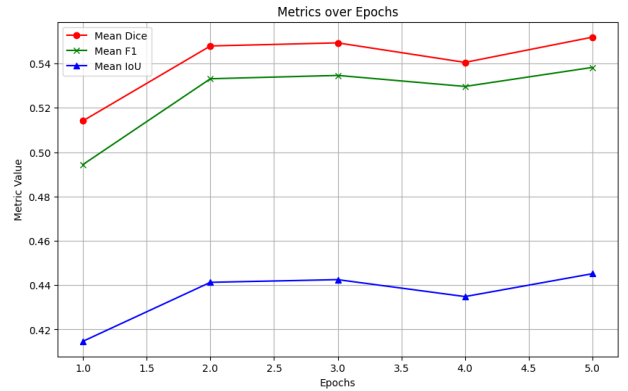


Fig. 3. Mean Dice, Mean F1, and Mean IoU Parameter Scores Across Epochs for Mask2Former Model

A. Variation of Number of Epochs

We can see the Loss Gradually Reducing and Converging to a value Quickly. If it was Run for a greater number of Epochs, Due to Over-shooting we Loose the Optimal point.

The overall increase in Mean Dice, Mean IoU, and Mean F1 scores, although with slight fluctuation reflects a good balance in the model's segmentation performance. The metric indicates a good overlap between the predicted and actual segmentation areas. The above Data is in Tabulated Format Below.

TABLE I
SEGMENTATION PERFORMANCE METRICS

Epoch	Mean Dice	Mean F1	Mean IoU
1	0.514	0.497	0.416
2	0.548	0.537	0.442
3	0.550	0.535	0.444
4	0.540	0.529	0.435
5	0.552	0.539	0.448

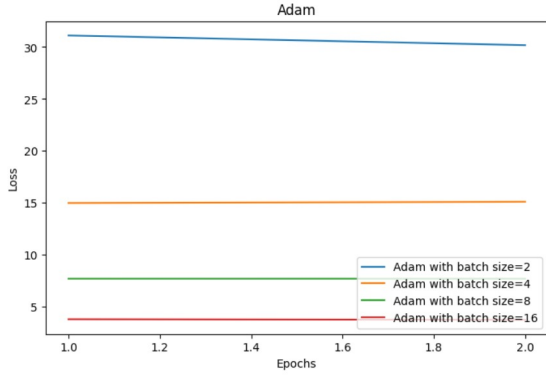


Fig. 4. Variation of Batch Size using Adam Optimizer Comparing Loss factor

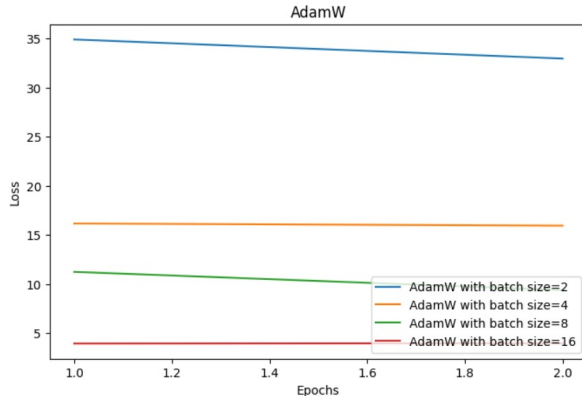


Fig. 5. Variation of Batch Size using AdamW Optimizer Comparing Loss factor

B. Variation of Batch size for different Optimizers

We observe the effect of varying batch sizes on the training dynamics of image segmentation models using Adam and AdamW optimizers. Larger batch sizes tend to stabilize training but may slow convergence, while smaller sizes accelerate the convergence. Comparing Adam and AdamW, we know that it is almost invariant changes for a small number of Epochs but by increasing the epochs we may observe differences between Optimizers.

VII. CONCLUSION

The Mask2Former model has proven to be effective in universal image segmentation tasks, and adapt to various challenging situations very well in the Indian road dataset. The model's concealed focus and high-resolution features enable accurate segmentation even in rarefied modes.

Future work will focus on further optimizing the Mask2Former architecture to improve small object segmentation, and explore the potential of advanced converter backbones for writing better encoding.

ACKNOWLEDGMENT

REFERENCES

- [1] Cheng, Bowen, Alexander G. Schwing, and Alexander Kirillov. *Per-Pixel Classification is Not All You Need for Semantic Segmentation*.

- In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4371-4380. 2021.
- [2] Cheng, Bowen, Ishan Misra, Alexander G. Schwing, and Alexander Kirillov. *Masked-attention Mask Transformer for Universal Image Segmentation*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1290-1299. 2022.
- [3] Hugging Face. *Transfer Learning with Transformers*. Available online: <https://colab.research.google.com/github/johko/computer-vision-course/blob/main/notebooks>
- [4] He, Kaiming, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. *Mask R-CNN*. In Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp. 2961-2969. 2017.
- [5] Liu, Ze, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. *Swin Transformer: Hierarchical Vision Transformer using Shifted Windows*. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 10012-10022. 2021.
- [6] Zhao, Hengshuang, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia. *Pyramid Scene Parsing Network*. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2881-2890. 2017.
- [7] Cordts, Marius, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. *The Cityscapes Dataset for Semantic Urban Scene Understanding*. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3213-3223. 2016.
- [8] Varma, M., Subramanian, V., Nambodiri, V., Chandraker, M., and Jawahar, C.V. *IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments*. Available online: <https://idd.insaan.iit.ac.in/dataset/details/> (accessed [Insert Date Accessed]).